YAZID AZZAOUI

M.Sc. Candidate in Computer Science · Northwestern Polytechnical University

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RESEARCH INTERESTS

Large language models · Generative AI and privacy · Multimodal learning for multi-omics · Biomedical data science

EDUCATION

M.Sc. in Computer Science 2024.09 – 2027.06
Northwestern Polytechnical University, Xi'an, China · Supervisor: Prof. Zhaowen Liu
Weighted average mark: 87.29 / 100
Thesis: "System-Level Framework for Interpretable Multi-Omic Network Modeling through LLM-Enhanced Knowledge Graphs"
Core courses: Algorithm Design and Analysis (95), Trustworthy Artificial Intelligence (93), High Performance Computing (92), Introduction to Bioinformatics (90)

B.Sc. in Software Engineering 2020.09 – 2024.06
Zhejiang University of Technology, Hangzhou, China
Graduation thesis: "Visual Analysis of Student Behavior in Learning Videos", Outstanding Graduation Thesis Award

RESEARCH EXPERIENCE

Master's thesis: LLM-enhanced knowledge graphs for multi-omics Ongoing
Northwestern Polytechnical University · Supervisor: Prof. Zhaowen Liu

- Developing a system-level framework that uses multimodal large language models to build and enrich biological knowledge graphs, for interpretable network modelling of multi-omic data.
- Predicting each of the nine PHQ-9 depression symptoms separately from proteomic, metabolomic, genetic and brain MRI data in UK Biobank, with every gain tested against permuted outcomes.

AEGIS: privacy-preserving generative model of disease trajectories 2026 – Present
Northwestern Polytechnical University · Supervisor: Prof. Zhaowen Liu · Role: first author

- Built and trained AEGIS, a GPT-style transformer of lifetime disease trajectories, on 251,204 UK Biobank participants with person-level differential privacy (DP-SGD, ε = 8, 3, 1).
- Designed a privacy audit with planted "canary" patients and membership-inference attacks, and evaluated how standard similarity checks for synthetic data behave.
- Released the code as open source (MIT licence, runs on simulated data): github.com/azzaouiyazid/AEGIS

Individualized proteomic network subtyping of major depressive disorder 2025 – 2026
NWPU · Shanghai Jiao Tong University · Fudan University · Role: co-first author

- Built an individualized network framework: a reference partial-correlation network of 2,911 plasma proteins from 37,054 controls, with per-patient deviations for 6,765 patients with MDD.
- Characterised the resulting depression subtypes and linked them to clinical outcomes with survival analysis (Kaplan–Meier, Cox models) and to brain MRI measures.

PUBLICATIONS

co-first author · * corresponding author

[1] **Yazid Azzaoui**, Xinran Wu, Xuequn Shang, Jie Zhang, Zhaowen Liu*. "AEGIS: private generative transformers that learn the natural history of disease, not the patients." Under review at *Nature Communications*. Code: github.com/azzaouiyazid/AEGIS

[2] **Yazid Azzaoui**#, Xinran Wu#, Zhengxu Lian, Xuequn Shang, Jie Zhang*, Zhaowen Liu*. "A complement-depleted axis in major depressive disorder, distinct from C-reactive protein: individualized plasma proteomic network analysis." Under review at *Brain, Behavior, and Immunity*.

CONFERENCES

[1] **Yazid Azzaoui**, Xinran Wu, Zhengxu Lian, Xuequn Shang, Jie Zhang, Zhaowen Liu. "Depression Is Not One Disease: Proteomic Links Between Cardiometabolic and Brainstem Pathways." Poster presentation, *OHBM 2026 Annual Meeting*, Bordeaux, France, June 2026.

[2] Participant, *2023 Apsara Conference* (Alibaba Cloud), "Computing for the value beyond computation", Hangzhou, China, 2023.

OTHER EXPERIENCE

Research apprentice in machine learning 2022.09 – 2023.01
Zhejiang University of Technology, Hangzhou, China

- Led a project classifying molecular bioactivity in PubChem with molecular fingerprints (MHFP) and neural networks.

Software engineering intern 2022.07 – 2022.10
F2MED, Casablanca, Morocco

- Built a web application for stock management with real-time tracking and analytics, and optimised the database behind it.

Founder and engineer (side project) Ongoing
CASABLANCAN LLC, USA

- Designed and built an online store from scratch: storefront, checkout and order tracking.

HONORS AND MEMBERSHIPS

- Chinese Government Scholarship (CSC), Master's program 2024 – 2027
- Student Member, Organization for Human Brain Mapping (OHBM) 2025 – present
- Outstanding Graduation Thesis Award, Zhejiang University of Technology 2024
- Zhejiang Provincial Scholarship 2023 – 2024

SKILLS

AI and deep learning	Transformers (GPT-style generative models), large language models (fine-tuning, prompting), PyTorch, TensorFlow, Hugging Face Transformers
Privacy and security	Differential privacy (DP-SGD), membership-inference attacks, privacy auditing, synthetic data evaluation
Statistics	Gaussian graphical models, survival analysis (Kaplan–Meier, Cox), permutation testing, clustering, enrichment analysis
Biomedical data	UK Biobank (Olink plasma proteomics, ICD-10 health records, brain MRI phenotypes), multi-omics integration
Programming	Python (NumPy, pandas, scikit-learn), R, SQL, Bash
Tools	Git, LaTeX, Linux GPU servers, full-stack web development

LANGUAGES

Arabic: fluent **French:** fluent **English:** fluent **German:** fluent **Chinese:** basic